



NCCER Module 22104



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Basic Operational Techniques





Section Two

*Starting, Operating, and Shutting Down
Heavy Equipment*

Objective

2. Describe the procedures for starting, operating, and shutting down heavy equipment.
 - a. Describe how to perform initial checks.
 - b. Describe basic operating procedures for the various items of equipment.
 - c. Explain how to properly shut down and secure equipment.



Performance Task

Under the supervision of an instructor, trainees should be able to do the following:

1. Perform basic prestart inspection, startup, operational movement, and shutdown for 3 of the 13 listed pieces of heavy equipment: utility tractors, dozers, loaders, backhoes, excavators, compaction equipment, motor graders, scrapers, on-road dump trucks, off-road dump trucks, forklifts, skid steers, trenchers



2.0.0 Starting, Operating, and Shutting Down Heavy Equipment

Certain actions must be taken before starting the engine. After startup, the engine needs to warm up for a short period before beginning operations. Refer to the O&M manual for specific amount of warmup time.



2.1.0 Initial Checks

Before starting the equipment, the operator must inspect it to verify that systems are working properly and that there is no physical damage to tires, wiring, hoses, and other equipment.



2.1.1 Indicator Checks

- Before starting the engine, turn the key to the On position and check the instrument panel, but do not start the engine.
- Make sure that all gauges and indicators function in accordance with the operator's manual.
- The instrument panel should light for a few seconds.
- If an indicator or gauge does not light up, check the bulb. Repeat the test.
- If the indicator still fails to light, have the machine serviced.



2.1.2 Starting Aids

- **Glow Plugs**—Prestart aids are often used with diesel engines. A current supplied to each glow plug causes it to produce heat. When the switch is activated, the plugs glow. When they stop glowing, the engine is warm enough to start.
- **Block Heaters**—An engine block heater is installed in the engine block water jacket. Typically, the heater is plugged into a standard electrical outlet.



2.1.3 Starting

- Verify that all controls are properly set before starting the machine.
- Check that the parking brakes are properly set.
- Start the machine only after completing any pre-starting procedures and fastening the seat belt.
- Move the gearshift to neutral.
- Set the engine control speed and choke as directed in the operator's manual.
- Sound the horn to alert people nearby.
- Turn the key to the start position and/or depress the start button.



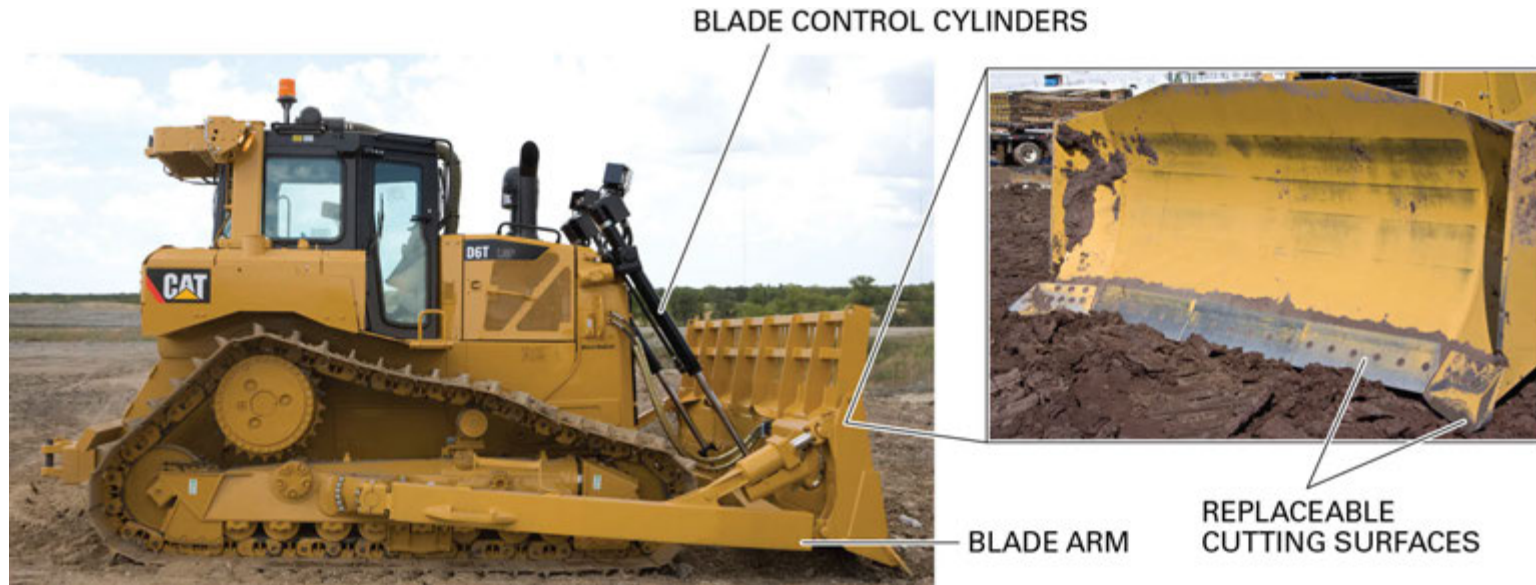
2.1.4 Warmup

- Operating the machine without warming it up can damage it.
- Run the machine for the time listed in the O&M manual.
- When the engine is running, check that all of the gauges and instruments are working properly.
- Circulate warm hydraulic oil throughout the machine by cycling the hydraulic controls.



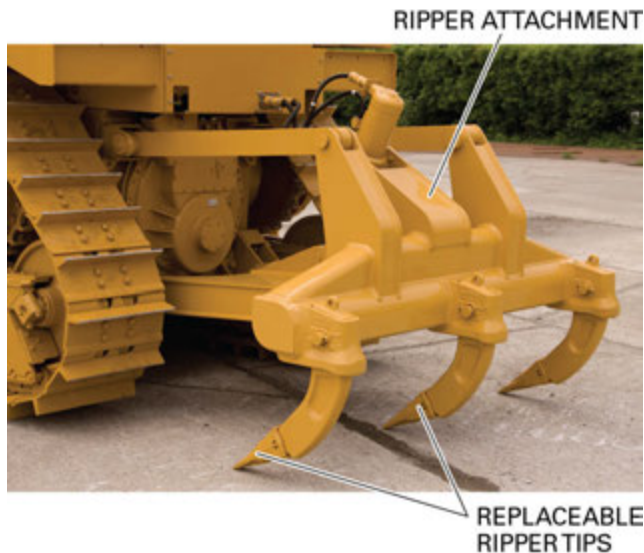
2.1.5 Attachments (1 of 3)

The primary attachment on dozers is the **blade**, along with the arms and cylinders supporting it.



2.1.5 Attachments (2 of 3)

- Dozers may have rippers.
- Graders have cutting blades called **moldboards**.



2.1.5 Attachments (3 of 3)

Loaders, backhoes, and excavators usually have buckets to dig or move materials such as dirt, rock, or waste products.



REPLACEABLE BUCKET TIPS

2.1.6 Operator Cab

The operator must start the machine and test-operate each control to verify that it functions correctly.



OPERATOR VIEW OUT FRONT
OF A JOHN DEERE BACKHOE



VIEW OF OPERATOR SEAT AND
CONTROLS FROM BACK OF A
JOHN DEERE BACKHOE

2.1.7 Prestart Activities for Trucks

- Prestart inspections are also necessary for dump trucks.
- Operators must review the O&M manual for each truck to become familiar with its controls.
- Complete any daily inspection forms.
- Note fluids added, service needed, and hours of operation.
- Make sure that the equipment is serviced regularly.



ON-ROAD DUMPTRUCK



ARTICULATED DUMP
TRUCK FOR UNDERGROUND MINING



RIGID-FRAME OFF-ROAD DUMPTRUCK

2.2.0 Equipment Operation

The way heavy equipment is built, and the way it steers makes a difference in how it is driven. The size of the machine must also be considered when it is operated. Tracked vehicles must be operated differently than wheeled vehicles.



2.2.1 Construction and Steering

- Most wheeled machines with straight frames have either a front or rear axle (or both) with wheels that can be turned to make the machine change directions.
- Other heavy equipment machines have hinged (articulated) frames.
- Tracked vehicles are steered by a lever or **joystick** that controls the speed and direction of the individual tracks.



HINGED AREA

2.2.2 Basic Operations

- Operators start training by learning basic maneuvers at a slow speed on level terrain.
- A trainer may set up an obstacle course for advanced maneuvering.
- The operator must be aware of what slope angles the machine can be safely operated on.



2.2.3 Machine-Specific Operations

Each machine operated by a heavy equipment operator has its own characteristics. The O&M manual for each machine lists any specific instructions on how to operate the machine.



2.2.4 Utility Tractors

- Steering utility tractors is similar to steering a personal vehicle.
- A tractor can flip over with or without an attachment. Tractors should not be driven on steep slopes.
- Operators must pay special attention to the hydraulic system used to operate attachments.



2.2.5 Backhoes

- The backhoe attachment has outriggers or load **stabilizers** that the operator extends to level and stabilize the machine as the backhoe attachment is used.
- The front bucket is lowered to raise the backhoe's front end while the backhoe attachment is being used.
- Raising the backhoe by its front bucket and its outriggers increases the risk of tipping over.



2.2.6 Skid Steers

Skid steers can be equipped with either wheels or tracks. Both types are steered by levers. They can be equipped with rotating tools.



WHEELED SKID STEER



SKID STEER WITH POST HOLE DIGGER



TRACKED SKID STEER



SKID STEER WITH SWEEPER BROOM

2.2.7 Loaders (1 of 2)

- Wheel loaders are the most common type. Track loaders are used in applications where the terrain requires them.
- Always travel with the mounted attachment in the stored position. For loaders with a bucket attachment, travel with the bucket lowered.



(A) WHEEL LOADER



(B) CRAWLER LOADER

2.2.7 Loaders (2 of 2)

- Loaders are steered by either a steering wheel or by levers depending on whether they are wheel loaders or track loaders.
- Loaders also have foot controls.
- Loader attachments are controlled by levers that operate the hydraulic systems used to drive the attachments.



OPERATOR AREA OF A
CATERPILLAR WHEELED LOADER



OPERATOR AREA OF A
CATERPILLAR TRACKED LOADER

2.2.8 Dozers (1 of 2)

- Dozers are usually equipped with a blade attachment on the front.
- Track dozers are steered by levers or joysticks.
- Wheel dozers are usually steered by steering wheels.
- Dozer blades are raised, lowered, or tilted by hydraulic system components controlled by levers in the operator's cab.



WHEEL DOZER



TRACK DOZER

2.2.8 Dozers (2 of 2)

- Brake before changing speed or direction.
- On older tracked machines, levers and pedals are used for turning.
- Newer machines have joystick controls for engine control and movement.
- Before operating a dozer, the operator must understand how to operate the blade and its controls.



2.2.9 Excavators

- Excavators can be equipped with buckets or other specialized attachments.
- They can be mounted on tracks or wheels.
- The boom and other attachments are controlled with joysticks or levers.



TRACKED EXCAVATOR EQUIPPED
WITH HAMMER ATTACHMENT



WHEELED EXCAVATOR EQUIPPED
WITH GRAPPLE ATTACHMENT

2.2.10 Scrapers

- Scrapers are highly mobile excavators used to dig, carry, and spread soil and other materials.
- Hydraulic systems are used to control their scraping and dumping functions.
- A steering wheel is used in combination with foot pedals to control vehicle movement.
- The joystick controls the moving components of the bowl.



TOWED SCRAPER



WHEELED TRACTOR SCRAPER

2.2.11 Motor Graders (1 of 2)

Motor graders are used to shape and finish materials.

They have four major components:

- The frame
- The power unit, including the engine, transmission, and differential
- The steering and other controls
- The circle, drawbar, moldboard, and blade



(A) RIGID FRAME MODEL



DRAWBAR AND CIRCLE
MOLDBOARD AND BLADE
(B) ARTICULATED FRAME MODEL

2.2.11 Motor Graders (2 of 2)

Primary operations that motor graders can perform with their blades:

- Rough grading
- Mixing rows of material
- Making rows of material for others to pick up (windrowing)
- Leveling new material
- Finish grading
- Ditch cutting or cleaning
- Light snow plowing

Scarifying, ripping, or heavy snow plowing can be done with attachments.



2.2.12 Compacting Equipment

Compactors are used to pack soil or gravel into a usable foundation. Specialized compactors are used for asphalt.



COMPACTOR PACKING SOIL



COMPACTOR PACKING GRAVEL



ASPHALT COMPACTOR WITH TWIN STEEL DRUMS



ASPHALT COMPACTOR WITH A STEEL DRUM AND RUBBER TIRES

2.2.13 Trenchers

- Trenching machines are used when narrow trenches are needed.
- Travel with the trenching bar and chain assembly raised.
- The trenching equipment is driven and controlled by a hydraulic system.
- Always check the trencher for loose or broken teeth.
- Check for roots, rocks, or other types of debris.



2.2.14 Dump Trucks

- A heavy-duty hydraulic system operates the bed.
- Newer trucks have camera systems that allow the driver to see what is behind the vehicle.
- Loaded trucks have a different center of gravity and cannot be stopped quickly.
- The four basic control positions are raise, hold, **float**, and lower.
- Never travel with the dump body raised.



RIGID-FRAMED TRUCK DUMPING



ARTICULATED TRUCK



ARTICULATED TRUCK
DUMPING THROUGH TAILGATE

2.2.15 Forklifts (1 of 2)

- Warehouse forklifts are built lower to the ground and operate on smaller, hard tires with smooth tread.
- Rough-terrain forklifts are built higher off the ground and operate on larger, treaded tires.
- The lifting forks are driven and controlled by a hydraulic system.



(A) WAREHOUSE FORKLIFT

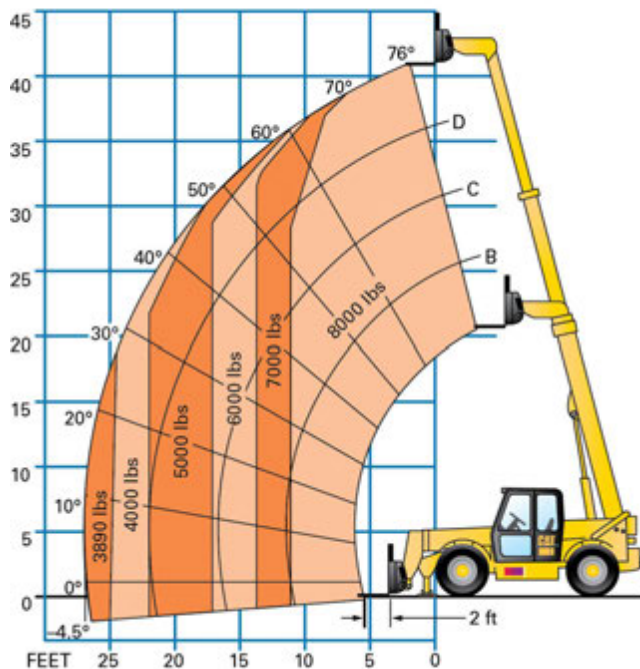


(B) ROUGH-TERRAIN FORKLIFT

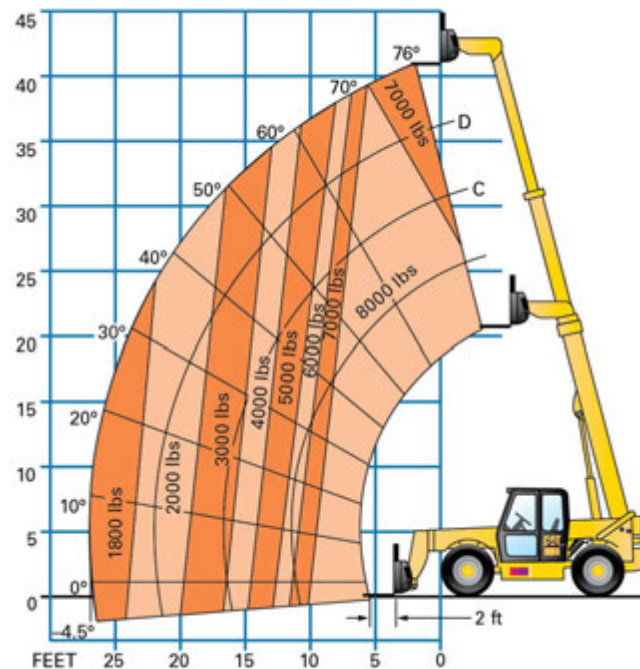
2.2.15 Forklifts (2 of 2)

The most important factor to consider when using a forklift is lifting capacity. Manufacturers supply a capacity chart with each forklift.

OPERATING SPECS



MACHINE WITH STABILIZERS DOWN



MACHINE NOT EQUIPPED WITH STABILIZERS

2.3.0 Shutdown Procedures

- Review the operator's manual for specific shutdown procedures.
- Park the machine on a level area.
- Lower the bucket or attachment.
- Depress the clutch and place all levers in neutral.
- Place the transmission in Neutral or Park.
- Engage the parking brake (if provided).
- Before turning the engine off, let it cool down.



Wrap Up – Trade Terms (1 of 2)

Blade: The primary attachment on a bulldozer used to push material in front of the equipment. It is typically a concave metal plate.

Float: A control setting where the bucket, forks, or blade will follow the ground contour.

Glow plugs: Heating elements used to preheat the combustion chamber and aid in igniting fuel in a cold engine.

Joystick: A control mechanism that pivots about a fixed point in four directions. It is used to control the motion of an object.

Moldboard: The cutting blade on a motor grader.

Pitch: The angle of the blade and/or moldboard in relation to a vertical plane.



Wrap Up – Trade Terms (2 of 2)

Power takeoff (PTO): A mechanism attached to a motor vehicle engine that supplies power to an off-vehicular device, such as an auger or pneumatic hammer.

Scarifying: Loosening the top surface of material using a set of metal shanks (teeth).

Stabilizers: Hydraulic arms on backhoes and other equipment that can be positioned on the right and left sides of the machine to give it additional stability.

Throttle: A lever that regulates the supply of fuel to an engine.

2.0.0 Section Review Question 1

1. Glow plugs are used to _____.
 - a. provide temporary lighting for the instrument panel
 - b. preheat the combustion chamber of a diesel engine
 - c. ignite the air and gasoline mixture in a gasoline engine
 - d. provide light for equipment inspection



2.0.0 Section Review Question 1 Answer

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2.0.0 Section Review Question 2

2. To turn a dozer in a new direction, you would _____.
- a. turn the steering wheel in the new direction
 - b. reverse one track while accelerating the other
 - c. brake one track while powering the other
 - d. lower one corner of the blade toward the turn direction



2.0.0 Section Review Question 2 Answer

2. To turn a dozer in a new direction, you would _____.
a. turn the steering wheel in the new direction
b. reverse one track while accelerating the other
c. brake one track while powering the other
d. lower one corner of the blade toward the turn direction



2.0.0 Section Review Question 3

3. Which of these steps would you NOT take when shutting down a loader?
- a. Place the gear shift in park or neutral.
 - b. Raise the bucket to its highest position.
 - c. Allow the engine to cool before turning it off.
 - d. Operate levers to release hydraulic pressure.



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Next...



Review and Module Examination

Complete the Review Questions at the end of the module and prepare for the Module Exam.